

Instrukcije Asemblera

LV3

EQU i ORG

- EQU → dodjeljivanje vrijednosti varijabli
- ORG → označava početak programa (zamjena za NOP)
 - Provjeri disassembly...

- Primjer

```
.equ varA=125
```

```
.equ varB=33
```

```
.equ varC=-33
```

```
.org 0x001D
```

```
inc R16
```

Aritmetičke operacije

- Sum
 - ADD, ADC
 - ADIW
 - INC
- Sub
 - SUB, SUBI
 - SBC, SBCI
 - SBIW
 - DEC
- Multiplication
 - MUL, MULS, MULSU
 - (FMUL, FMULS, FMULSU)
- Logic
 - AND, ANDI
 - OR, ORI
 - EOR
 - COM
 - NEG
- Bit Manipulation
 - SBR, CBR
 - SER, CLR
- Special
 - TST

- ADD

Description

Operation:

Syntax:

Operands:

Program Counter:

(i) ADD Rd,Rr

$$0 \leq d \leq 31, 0 \leq r \leq 31$$
$$PC \leftarrow PC + 1$$

16-bit Opcode:

0000	11rd	dddd	rrrr
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
-	-	↔	↔	↔	↔	↔	↔

Primjer

```
ldi r16,5      ; load 5 into register 16  
ldi r17,6      ; load 6 into register 17  
add r16,r17    ; add r16 and r17 and store the result in r16
```

- ADC

Description

Operation:

Syntax:

Operands:

Program Counter:

16-bit Opcode:

Status Register (SREG) and Boolean Formula

[illegible]

Primjer

```
ldi r16,0x34    ; place lower byte of 0x1234 in r16  
ldi r17,0x12    ; place upper byte of 0x1234 in r17  
ldi r18,0xCD    ; place lower byte of 0xABCD in r18  
ldi r19,0xAB    ; place upper byte of 0xABCD in r19
```

```
add r16,r18 ; compute sum of lower bytes (result = 0x01)  
adc r17,r19 ; compute sum of upper bytes with carry  
              (result = 0xBE)
```

SUM

- ADIW

ADIW – Add Immediate to Word

Description

Adds an immediate value (0 - 63) to a register pair and places the result in the register pair. This instruction operates on the upper four register pairs, and is well suited for operations on the pointer registers.

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

(i) $Rd+1:Rd \leftarrow Rd+1:Rd + K$

Syntax:

Operands:

Program Counter:

(i) ADIW Rd+1:Rd,K

$d \in \{24,26,28,30\}, 0 \leq K \leq 63$

$PC \leftarrow PC + 1$

16-bit Opcode:

1001	0110	KKdd	KKKK
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$

Primjer 1.

```
ldi r24,0x00    ; load 0x1000 into  
ldi r25,0x10    ; registers r24:r25  
adiw r24,0x0A   ; add 0x0A to r24:r25 (result = 0x100A)  
ldi XL,0x80     ; load 0x8080 into  
ldi XH,0x80     ; X pointer  
adiw X,1        ; increment X pointer (result = 0x80801)
```

Upravljanje memorijom - adresiranje

SUM

- INC

INC – Increment

Description

Adds one -1- to the contents of register Rd and places the result in the destination register Rd.

The C Flag in SREG is not affected by the operation, thus allowing the INC instruction to be used on a loop counter in multiple-precision computations.

When operating on unsigned numbers, only BREQ and BRNE branches can be expected to perform consistently. When operating on two's complement values, all signed branches are available.

Operation:

(i) $Rd \leftarrow Rd + 1$

Syntax:

(i) INC Rd

Operands:

$0 \leq d \leq 31$

Program Counter:

$PC \leftarrow PC + 1$

16-bit Opcode:

1001	010d	dddd	0011
------	------	------	------

Status Register and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	–

Zadatak 1.

- Napiši program koji izvršava sljedeće jednačbe

$$Y1 = \text{varA} + \text{varB}$$

$$Y2 = \text{varA} + \text{varC}$$

$$Y3 = \text{varB} + \text{varC}$$

Rezutate sprema na sljedeće adrese u memoriji

$$Y1=64, Y2=65, Y3=66$$

Za provjeru koristite sljedeće vrijednosti

- $\text{varA} = 125, \text{varB} = 35, \text{varC} = -33.$

- SUB

Description

Operation:

(i) $R_d \leftarrow R_d - R_r$

Program Counter:

$$PC \leftarrow PC + 1$$

16-bit Opcode:

0001	10rd	dddd	rrrr
------	------	------	------

I	T	H	S	V	N	Z	C
-	-	↔	↔	↔	↔	↔	↔

Primjer

```
ldi r16,0x01    ; place byte of 0xBE01 in r16  
ldi r18,0x34    ; place byte of 0x1234 in r18  
  
sub r16,r18      ; subtract bytes (result = 0xCD)
```

- SUBI

Description

Operation:

(i) $R_d \leftarrow R_d - K$

Operands:

Program Counter:

(i) SUBI Rd,K

$$16 \leq d \leq 31, 0 \leq K \leq 255$$

$$PC \leftarrow PC + 1$$

16-bit Opcode:

0101	KKKK	dddd	KKKK
------	------	------	------

I	T	H	S	V	N	Z	C
-	-	↔	↔	↔	↔	↔	↔

Primjer

ldi r16,0x05 ; load 0x05 into register 16

sbi r16,0x05 ; subtract 5 from r16 (result = 0x00)

SUB

SBC – Subtract with Carry

Description

Subtracts two registers and subtracts with the C Flag, and places the result in the destination register Rd.

- SBC

Operation:

(i) $R_d \leftarrow R_d - R_r - C$

Syntax:

Operands:

Program Counter:

(i) SBC R_d, R_r

$$0 \leq d \leq 31, 0 \leq r \leq 31$$

$$PC \leftarrow PC + 1$$

16-bit Opcode:

0000	10rd	dddd	rrrr
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
-	-	↔	↔	↔	↔	↔	↔

Primjer

ldi r16,0x01 ; place lower byte of 0xBE01 in r16

ldi r17,0xBE ; place upper byte of 0xBE01 in r17

ldi r18,0x34 ; place lower byte of 0x1234 in r18

ldi r19,0x12 ; place upper byte of 0x1234 in r19

sub r16,r18 ; subtract lower bytes (result = 0xCD)

sbc r17,r19 ; subtract upper bytes with carry (result = 0xAB)

SUB

- SBCI

Operation:

Syntax:

Program Counter:

$$PC \leftarrow PC + 1$$

0100	KKKK	dddd	KKKK
------	------	------	------

I	T	H	S	V	N	Z	C
-	-	↔	↔	↔	↔	↔	↔

SBIW – Subtract Immediate from Word

SUB

- SBIW

Description

Subtracts an immediate value (0-63) from a register pair and places the result in the register pair. This instruction operates on the upper four register pairs, and is well suited for operations on the Pointer Registers.

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

(i) $Rd+1:Rd \leftarrow Rd+1:Rd - K$

Syntax:

Operands:

Program Counter:

(i) SBIW Rd+1:Rd,K $d \in \{24,26,28,30\}, 0 \leq K \leq 63$ $PC \leftarrow PC + 1$

16-bit Opcode:

1001	0111	KKdd	KKKK
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	↔	↔	↔	↔	↔

Primjer

ldi YL,0xAA ; load 0x55AA into

ldi YL,0x55 ; Y pointer

sbiw Y,0x10 ; subtract 0x10 from Y (result = 0x559A)

DEC – Decrement

SUB

- DEC

Description

Subtracts one -1- from the contents of register Rd and places the result in the destination register Rd.

The C Flag in SREG is not affected by the operation, thus allowing the DEC instruction to be used on a loop counter in multiple-precision computations.

When operating on unsigned values, only BREQ and BRNE branches can be expected to perform consistently. When operating on two's complement values, all signed branches are available.

Operation:

(i) $Rd \leftarrow Rd - 1$

Syntax:

(i) DEC Rd

Operands:

$0 \leq d \leq 31$

Program Counter:

$PC \leftarrow PC + 1$

16-bit Opcode:

1001	010d	dddd	1010
------	------	------	------

Status Register and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	–

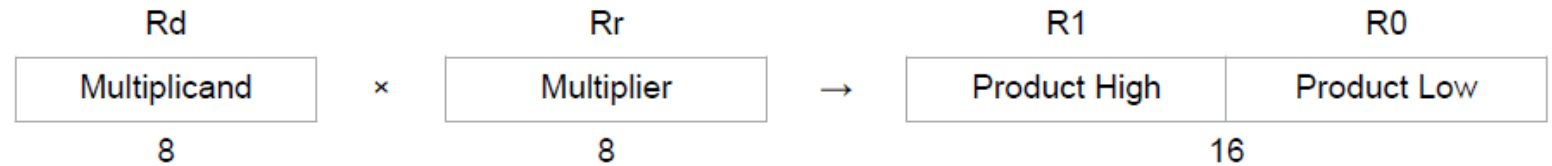
Multiplication

- MUL (pozitivne vrijednosti)

MUL – Multiply Unsigned

Description

This instruction performs 8-bit $\times$ 8-bit $\rightarrow$ 16-bit unsigned multiplication.



The multiplicand Rd and the multiplier Rr are two registers containing unsigned numbers. The 16-bit unsigned product is placed in R1 (high byte) and R0 (low byte). Note that if the multiplicand or the multiplier is selected from R0 or R1 the result will overwrite those after multiplication.

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

- (i) $R1:R0 \leftarrow Rd \times Rr$ (unsigned $\leftarrow$ unsigned $\times$ unsigned)

Syntax:

- (i) MUL Rd,Rr

Operands:

$0 \leq d \leq 31, 0 \leq r \leq 31$

Program Counter:

$PC \leftarrow PC + 1$

16-bit Opcode:

1001	11rd	dddd	rrrr
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	$\Leftrightarrow$	$\Leftrightarrow$

Primjer

```
ldi r16,0x18    ; load r16 with 0x18 (22)
ldi r17,0x37    ; load r17 with 0x37 (55)
mul r16,r17     ; multiply r16 and r17 (r1:r0 = 0x0528 = 1210)
```

MULS – Multiply Signed

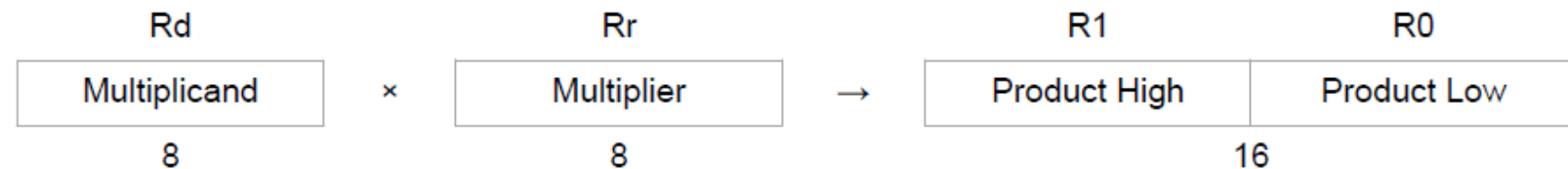
Multiplication

- MULS

- Signed (pozitivne i negativne vrijednosti)

Description

This instruction performs 8-bit × 8-bit → 16-bit signed multiplication.



The multiplicand Rd and the multiplier Rr are two registers containing signed numbers. The 16-bit signed product is placed in R1 (high byte) and R0 (low byte).

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

- (i) $R1:R0 \leftarrow Rd \times Rr$ (signed ← signed × signed)

Syntax:

Operands:

Program Counter:

- (i) MULS Rd,Rr

$16 \leq d \leq 31, 16 \leq r \leq 31$

$PC \leftarrow PC + 1$

16-bit Opcode:

0000	0010	dddd	rrrr
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	↔	↔

Primjer

```
ldi r16,0x18    ; load r16 with 0x18 (22)
ldi r17,0x37    ; load r17 with 0x37 (55)
mul r16,r17     ; multiply r16 and r17 (r1:r0 = 0x0528 = 1210)
```

Primjer

```
ldi r16,0xF2    ; load r16 with 0xF2 (-14)
ldi r17,0xDF    ; load r17 with 0xDF (-33)
mul r16,r17     ; multiply r16 and r17 (r1:r0 = 0x01CE=462)
```

MULSU – Multiply Signed with Unsigned

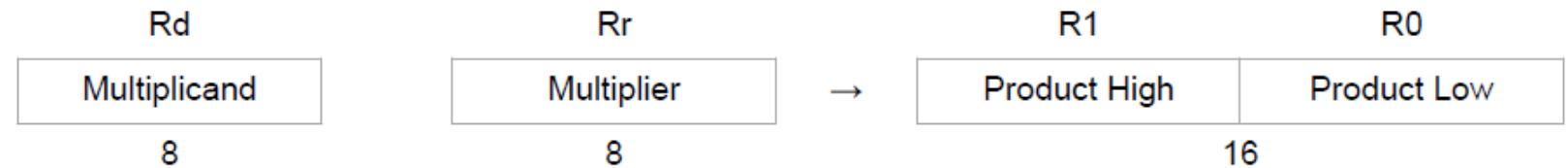
Multiplication

- MULSU

- Umnožak “signed”
i “unsigned” broja

Description

This instruction performs 8-bit × 8-bit → 16-bit multiplication of a signed and an unsigned number.



The multiplicand Rd and the multiplier Rr are two registers. The multiplicand Rd is a signed number, and the multiplier Rr is unsigned. The 16-bit signed product is placed in R1 (high byte) and R0 (low byte).

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

- (i) $R1:R0 \leftarrow Rd \times Rr$ (signed ← signed × unsigned)

Syntax:

Operands:

Program Counter:

- (i) MULSU Rd,Rr

$16 \leq d \leq 23, 16 \leq r \leq 23$

$PC \leftarrow PC + 1$

16-bit Opcode:

0000	0011	0ddd	0rrr
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	↔	↔

Primjer

ldi r16,0xF2 ; load r16 with 0xF2 (-14)

ldi r17,0x7D ; load r17 with 0x83 (125)

mulsu r16,r17 ; multiply r16 and r17 (r1:r0 = 0xF92A = -1750)

Zadatak 2.

- Napiši program koji izvršava sljedeću jednadžbu

$$Y = ((A * B + C) * A - (C * A)) * (B * A + C - B)$$

Rezultat sprema na sljedeću adresu u memoriji
adresa 64

Za provjeru koristite sljedeće vrijednosti

- $A = 3$, $B = 5$, $C = -7$.

Varijable, međurezultati i rezultati stanu u 8-bitni registar